

Separate input from output

An RAPD reflects unequal afferent input; a fixed single pupil can reflect an output problem.

Interpret the pupil pattern

Observation	Inference
Both pupils relatively dilate when light enters the left eye	Left afferent input is weaker.
Right pupil fails; left constricts with either eye illuminated	Right efferent or iris response is impaired.
Equal resting pupil sizes	An RAPD is still possible.
Unequal resting pupil sizes	Assess the efferent pathway or iris separately.

Use the paired responses

Direct and consensual constriction let the two afferent inputs be compared through the same pupil outputs. Do not use resting size alone to exclude an afferent problem. Retinal disease affecting a large area or optic-nerve disease may impair afferent input; the pupil finding needs interpretation alongside acuity, colour, fields and the fundus.

Test relative function reproducibly

A distant fixation target and equal light presentations reduce misleading pupil responses.

Afferent screening sequence

Check resting pupil size and the direct and consensual responses.



Keep distant fixation; illuminate one eye for about three seconds, then the other.



Look for relative dilation when light reaches the eye with weaker input.



If that pupil is fixed, observe the reactive fellow pupil for a reverse RAPD.

Add colour quality

Compare a red object with each eye separately. Use the better eye's red as the reference and ask whether the other eye sees it less vividly. Ishihara plates assess colour recognition, while red-saturation comparison can reveal an asymmetry in perceived intensity. These findings complement, rather than replace, the chart acuity and pupil assessment.